

CASE STUDY 1: Identity Management Failure in Cloud System

A multinational company migrates its applications to the cloud but continues using weak password-based authentication without multi-factor authentication (MFA). Several user accounts are compromised, leading to data breaches.

Questions:

1. Analyze the weaknesses in the company's identity management system.
2. Explain how passwords, tokens, and biometrics could improve security.
3. Design a secure identity management solution for this system.
4. Discuss the role of Single Sign-On (SSO) in this scenario and its risks.

CASE STUDY 2: Secure Execution Environment Breach

A cloud provider hosts multiple tenants on shared virtual machines. One tenant exploits a vulnerability and gains access to another tenant's data.

Questions:

1. Explain how secure execution environments should prevent such attacks.
2. Identify architectural flaws in the system.
3. Suggest hardware-based and software-based protection mechanisms.
4. Discuss how isolation and virtualization security can be improved.

CASE STUDY 3: Access Control Misconfiguration

An organization uses Role-Based Access Control (RBAC) but assigns excessive permissions to users, violating the principle of least privilege.

Questions:

1. Compare different access control models (DAC, MAC, RBAC).
2. Analyze the risks of improper RBAC implementation.
3. Propose a corrected access control policy.
4. Explain how identity management integrates with access control.

CASE STUDY 4: Intrusion Detection Failure in Cloud

A company relies on an Intrusion Detection System (IDS), but it fails to detect a slow data exfiltration attack.

Questions:

1. Explain layered security and the role of IDS in cloud environments.

2. Identify limitations/issues of IDS in cloud computing.
3. Propose improvements using advanced monitoring techniques.
4. Design a multi-layered security architecture to prevent such attacks.

CASE STUDY 5: Encryption and Key Management Issue

A cloud service provider stores encryption keys along with encrypted data, leading to a security breach when the server is compromised.

Questions:

1. Explain the importance of key management in cloud security.
2. Identify the mistake in this implementation.
3. Suggest a secure key management architecture.
4. Discuss encryption standards and best practices for cloud data protection.

CASE STUDY 6 : AWS IAM Misconfiguration

An organization uses AWS Identity and Access Management (IAM), but improper policy configuration leads to unauthorized access.

Questions:

1. Explain AWS IAM components (users, roles, policies).
2. Identify common IAM misconfigurations.
3. Design a secure IAM policy using least privilege principle.
4. Discuss best practices for AWS IAM security.

CASE STUDY 7 : Hardware vs Software Security

A cloud provider relies only on software-based security mechanisms, ignoring hardware-level protections.

Questions:

1. Compare hardware-based and software-based protection.
2. Explain Trusted Platform Module (TPM) and secure boot.
3. Analyze risks of ignoring hardware security.
4. Suggest a combined security architecture.